

# ELIE GAMBACHE

## Kotlin Multiplatform Engineer — Compose Desktop & Native Systems Architecture

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### PROFESSIONAL SUMMARY

**Kotlin Multiplatform** specialist and systems architect focused on native desktop development. Creator of **Nucleus**, the native desktop platform for the JVM, and author of **30+ open-source Kotlin libraries** (2,900+ combined GitHub stars) trusted by JetBrains and ETH Zürich and shipping in apps with 280K+ downloads. Expert at writing clean **Kotlin** over low-level OS APIs and bridging Rust, C++ and Go into the JVM.

### PROFESSIONAL EXPERIENCE

**Founder & Lead Open-Source Engineer** — *KdroidFilter / NucleusFramework, Jerusalem* **2023 – Present**

- Created **Nucleus** — the native desktop platform for the JVM: a Gradle-plugin framework (30+ modules, published to Maven Central) delivering native windows, OS integration, code signing and installers from a single DSL, on GraalVM native image or the JVM.
- Authored **30+ open-source Kotlin Multiplatform libraries** for the Compose ecosystem — native media playback, system tray, WebView, notifications, accessibility and more.
- Key Achievement:** libraries received open-source support from **JetBrains**, were **sponsored by ETH Zürich**, and now ship in production apps totalling **280,000+ downloads** (Pushscroll, AeroDL, AB Download Manager — 12K★).
- Key Achievement:** ranked **#6 GitHub committer in Israel** (committers.top) with 2,900+ stars earned across personal and organization repositories.

### KEY PROJECTS

**Nucleus** **★ 304 · Maven Central**

- Native desktop platform for the JVM — Compose Multiplatform + native window decorations, OS integration, code signing, notarization; ~0.5 s cold start, 60 MB RAM; signed installers in 16 formats from a git tag.

**ComposeMediaPlayer** **★ 475**

- The only Compose media player using native OS codecs (MediaFoundation, AVPlayer, GStreamer, ExoPlayer) — no VLC/MPV, zero external dependencies. Supported by JetBrains; used by Pushscroll (280K+).

**ComposeNativeTray** **★ 398**

- True native system tray for Compose Desktop (Go + D-Bus, C++, native macOS). Sponsored by ETH Zürich; used by AB Download Manager (12K★).

**ComposePdfReader** **★ 59**

- KMP PDF rendering & text extraction built on pdfium — Compose-first API with progressive rendering and selectable text; sample reader for desktop and mobile.

**Zayit** **~15,000 downloads**

- Production KMP app: semantic, fully-offline search over 7,000+ Jewish texts — Hebrew-aware tokenization, full-text indexing, citation-aware navigation on SQLite. Powered by Sefaria.

### EDUCATION

Self-taught software engineer — expertise built entirely through open-source practice, shipping production libraries and applications adopted across the industry.

### COMPUTER SKILLS

|                  |                                                                                                |
|------------------|------------------------------------------------------------------------------------------------|
| Languages        | Kotlin — primary language (expert)                                                             |
| Frameworks       | Compose Multiplatform, Kotlin Multiplatform (KMP)                                              |
| Native bridging  | JNI, Java FFM, JNA, UniFFI, GraalVM native image                                               |
| Platform APIs    | Win32 / WinRT, AppKit / AVFoundation, D-Bus, MediaFoundation, GStreamer                        |
| Build & delivery | Gradle plugin development, GitHub Actions, code signing, notarization, installers, auto-update |
| Domains          | Android MDM & enterprise mobility, search engines & data pipelines, Hebrew NLP                 |